

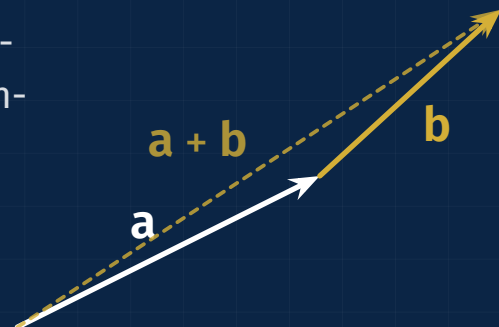


DR ESLAM AHMED
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CHAPTER 5 • VECTORS & TRANSFORMATIONS • TOPIC 2

Transformations

Translate, reflect, rotate, enlarge. To draw one, move each corner; to describe one, give every detail — vector, mirror line, angle + direction + centre, scale factor + centre.



5

VARIANTS

18

PAST-PAPER PARTS

33

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Translations

V2 Reflections

V3 Rotations

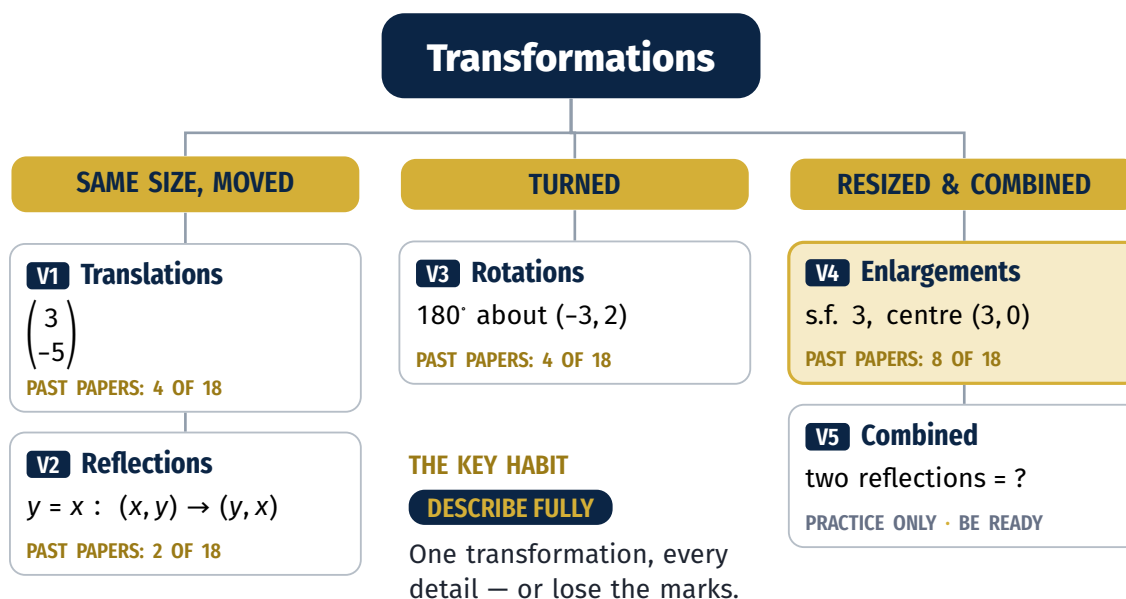
V4 Enlargements

V5 Combined transformations

TOPIC MAP · EVERY VARIANT ON ONE PAGE

Transformations at a glance



Your notes


How often does each variant come up?

9 past-paper questions (Jan 2020 → Jun 2025) hold 18 parts; 33 practice parts — grouped by type.

	PAST PAPERS (OUT OF 18)	PRACTICE PARTS
V1 Translations	4	10
V2 Reflections	2	5
V3 Rotations	4	7
V4 Enlargements	8	7
V5 Combined	0	4

WHAT THIS TELLS YOU Early in the paper (Q1–Q10, 2–5 marks), often two parts: draw one transformation, describe another. Enlargements (with fractional scale factors) are the favourite.

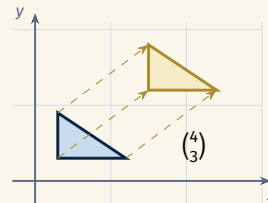
BEFORE YOU START

The big idea

**Move the corners, then join them.**

Every transformation moves each **vertex** by the same rule. Do the corners one at a time, then join them in the same order.

Translations, reflections and rotations keep the **size**; enlargements change it.

**“Describe fully” needs****Translation**

$$\begin{pmatrix} x \\ y \end{pmatrix}$$

The column vector.

Reflection

line

Its equation:
 $x = 2$, $y = x$.

Rotation

θ , dir, (a, b)

Angle, direction,
centre.

Enlargement

k , (a, b)

Scale factor and
centre.

**Quick rules about the origin**

In $y = x$

$$(x, y) \rightarrow (y, x)$$

Swap.

In $y = -x$

$$(x, y) \rightarrow (-y, -x)$$

Swap and negate.

90° clockwise

$$(x, y) \rightarrow (y, -x)$$

About O.

180°

$$(x, y) \rightarrow (-x, -y)$$

About O.

**CALCULATOR FIRST, THEN CHECK**

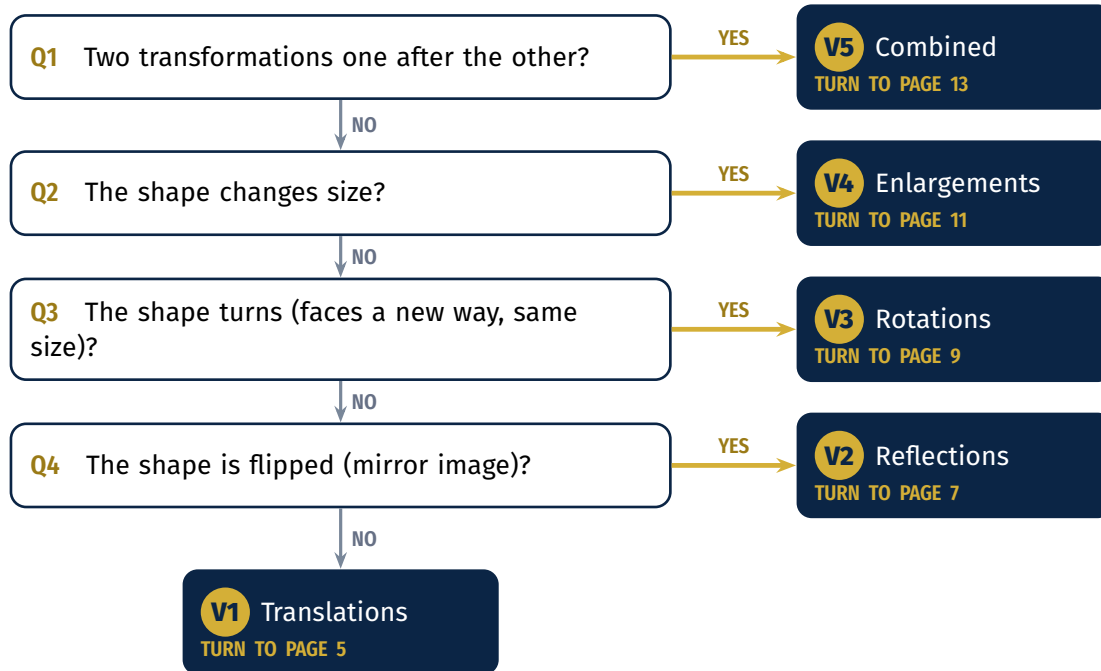
Tracing paper is allowed — use it for every rotation. Check one corner with the rule.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1

VARIANT 1

Translations

Every point moves by the same vector:
 $\begin{pmatrix} \text{right} \\ \text{up} \end{pmatrix}$.

SEEN IN PAST PAPERS

4 / 18

PRACTICE QUESTIONS: 10

 Your notes

Came up in: Jan 2021 P2HR Q5(b) • Jun 2022 P2HR Q1(b) • Jun 2024 P2HR Q2(a) • May 2023 P1H Q4(a)

? What is it?

In plain words

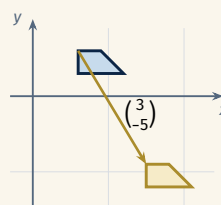
A translation **slides** the shape. $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$: 3 right, 5 down. The shape does not turn or change size.

To find the vector: pick one corner and its image; subtract:

image - start.

IT LOOKS LIKE

$\begin{pmatrix} -4 \\ 3 \end{pmatrix}$ find d e f



How do I do it?

1 Vector

Top: right (+) / left (-). Bottom: up / down.

2 Each corner

Add the vector.

3 Join

Same order; label the new shape.

4 Describe

"Translation by the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ ".

IMAGE - START Gives the vector.

V1 Translations

VARIANT 1 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

“translate triangle A by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$ ” → **Each corner: 4 left, 3 up**

“Describe fully the single transformation” (same way up, same size) → **Translation $\begin{pmatrix} a \\ b \end{pmatrix}$**

Coordinates with letters d, e, f → **Same vector for every corner**



WORKED EXAMPLE PAST PAPER

May 2023 · Paper 1H · Q4(a) · 3 marks

Triangle A: $(-7, 2)$, $(-11, -5)$, $(f, -3)$ is translated to triangle B: $(6, 12)$, $(2, d)$, $(9, e)$. Find d, e and f.

STEP 1 $(-7, 2) \rightarrow (6, 12) : \begin{pmatrix} 13 \\ 10 \end{pmatrix}$ ◀ the vector

STEP 2 $-5 + 10 = d \Rightarrow d = 5$ ◀ $(-11, -5) \rightarrow (2, d)$

STEP 3 $f + 13 = 9 \Rightarrow f = -4$ ◀ x of the third corner

STEP 4 $-3 + 10 = e \Rightarrow e = 7$ ◀ y of the third corner

ANSWER $d = 5, e = 7, f = -4$

✓ $-11 + 13 = 2$ ✓



WATCH OUT

✗ “moved 3 right and 5 down”

✓ “translation $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$ ”

Use the word and the column vector.



EXAM TIP

Check the vector on a second corner before you write it.



Check yourself

calculator first · answers on page 15

1.1

Jun 2024 P2HR

A: $(2, 1)$, $(2, 2)$, $(3, 2)$, $(4, 1)$. B: $(5, -4)$, $(5, -3)$, $(6, -3)$, $(7, -4)$. Describe $A \rightarrow B$.

Answer:

1.2

Jun 2022 P2HR

A: $(5, 1)$, $(5, 3)$, $(6, 1)$. Translate A by $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$.

Answer:

1.3

Dr Eslam

Translate $(1, 2)$, $(4, 2)$, $(1, 5)$ by $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$.

Answer:

V2

VARIANT 2

Reflections

Each corner the same distance on the other side of the mirror line, measured at right angles.

SEEN IN PAST PAPERS

2 / 18

PRACTICE QUESTIONS: 5


 Your notes

Came up in: Jun 2025 P2HR Q5(a) • Nov 2023 P2H Q3(a)

? What is it?

In plain words

Draw the **mirror line** first. Each corner goes straight across it (at 90°) to the **same distance** on the other side.

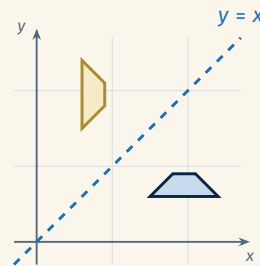
In $y = x$: **swap** the coordinates. $x = a$ lines are vertical; $y = b$ lines are horizontal.

IT LOOKS LIKE

$y = x$

$x = 2$

$y = -1$



How do I do it?

1 Mirror

Draw the line ($x = 2$, $y = x \dots$).

2 Distance

Each corner's distance to the line, at 90° .

3 Other side

Same distance across.

4 Describe

"Reflection in the line $y = x$ ".

SAME DISTANCE, OTHER SIDE $y = x$: swap x and y .

V2 Reflections

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Reflect shape A in the line $y = x$ " → $(x, y) \rightarrow (y, x)$ for each corner

"in the line $x = 2$ " → Vertical mirror through $x = 2$

Image flipped, same size → Reflection: find the mirror line between them

WORKED EXAMPLE **PAST PAPER**

November 2023 · Paper 2H · Q3(a) · 2 marks

Shape T has corners (1, 5), (1, 6), (4, 6), (4, 5), (3, 5), (3, 4), (2, 4), (2, 5).
Reflect T in the line $y = x$.

STEP 1 $(x, y) \rightarrow (y, x)$ ◀ the rule for $y = x$

STEP 2 $(1, 5) \rightarrow (5, 1)$, $(1, 6) \rightarrow (6, 1)$ ◀ swap

STEP 3 $(4, 6) \rightarrow (6, 4)$, $(4, 5) \rightarrow (5, 4)$, $(3, 5) \rightarrow (5, 3)$ ◀ swap

STEP 4 $(3, 4) \rightarrow (4, 3)$, $(2, 4) \rightarrow (4, 2)$, $(2, 5) \rightarrow (5, 2)$ ◀ swap; join in order

ANSWER T on its side, corners (5, 1), (6, 1), (6, 4), ... ✓ same shape, mirrored ✓

WATCH OUT

✗ reflecting in the x-axis

✓ reflect in $y = x$ (the diagonal)

Read the equation of the mirror line.

EXAM TIP

Turn the page so the mirror line is vertical — reflections get easier.

Check yourself

calculator first · answers on page 15

2.1 Jun 2025 P2HR

A: (5, 2), (6, 3), (7, 3), (8, 2). Reflect A in $y = x$.

Answer:

2.2 Dr Eslam

Reflect the point (5, 1) in the line $x = 2$.

Answer:

2.3 Dr Eslam

Reflect the point (5, 1) in the line $y = -1$.

Answer:

V3

VARIANT 3

Rotations

Angle, direction, centre: turn each corner about the centre (tracing paper).

SEEN IN PAST PAPERS

4 / 18

PRACTICE QUESTIONS: 7



Your notes

Came up in: Jan 2021 P2HR Q5(a) · Jun 2024 P2HR Q2(b) · Jun 2025 P2H Q2(b) · May 2023 P1H Q4(c)

? What is it?

In plain words

A rotation **turns** the shape about a **centre**. 180° : each corner goes straight through the centre to the same distance beyond.

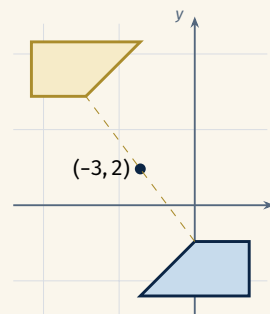
180° about (a, b) :

$(x, y) \rightarrow (2a - x, 2b - y)$.

IT LOOKS LIKE

180° about $(-3, 2)$

90° clockwise about $(3, 5)$



How do I do it?

1 Centre

Mark it.

2 Each corner

Turn it (tracing paper, or the rule).

3 Join

Same order.

4 Describe

"Rotation 90° clockwise about (a, b) ".

ANGLE · DIRECTION · CENTRE 180° needs no direction.

V3 Rotations

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“rotate shape P 180° about the point $(-3, 2)$ ” → $(x, y) \rightarrow (-6 - x, 4 - y)$

“rotate 90° clockwise about $(3, 5)$ ” → **Tracing paper; check one corner**

Image turned, same size: “describe fully” → **Find the centre: equal distances**

WORKED EXAMPLE **PAST PAPER**

January 2021 · Paper 2HR · Q5(a) · 2 marks

Shape P has corners $(0, -2)$, $(3, -2)$, $(3, -5)$, $(-3, -5)$. Rotate P 180° about $(-3, 2)$.

STEP 1 $(x, y) \rightarrow (-6 - x, 4 - y)$ ◀ 180° about $(-3, 2)$

STEP 2 $(0, -2) \rightarrow (-6, 6)$, $(3, -2) \rightarrow (-9, 6)$ ◀ first two corners

STEP 3 $(3, -5) \rightarrow (-9, 9)$, $(-3, -5) \rightarrow (-3, 9)$ ◀ last two

STEP 4 Join the four points: shape Q ◀ join

ANSWER $Q: (-6, 6), (-9, 6), (-9, 9), (-3, 9)$ ✓ $(-3, 2)$ is the midpoint of each pair ✓

WATCH OUT

✗ rotating about the origin

✓ about $(-3, 2)$

The centre changes everything.

EXAM TIP

Check: the centre is the midpoint of each corner and its image (for 180°).

Check yourself

calculator first · answers on page 15

3.1 May 2023 P1H

$P: (2, 5), (2, 3), (3, 2), (3, 3)$. Rotate P 90° clockwise about $(3, 5)$.

Answer:

3.2 Jun 2025 P2H

$P: (2, 2), (4, 2), (2, 6)$ maps to $Q: (8, 8), (6, 8), (8, 4)$. Describe fully.

Answer:

3.3 Dr Eslam

Rotate $(3, 1)$ through 90° clockwise about the origin.

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM**Enlargements**

Each corner k times as far from the centre:
 image = centre + k (corner – centre).

SEEN IN PAST PAPERS

8 / 18

PRACTICE QUESTIONS: 7



Your notes

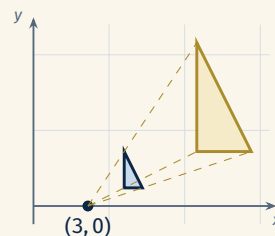
Came up in: Jan 2020 P1H Q8 · Jan 2021 P2HR Q5(c) · Jun 2022 P2HR Q1(a) · May 2023 P1HR Q10 · May 2023 P1H Q4(b) · Nov 2023 P2H Q3(b) · Jun 2025 P2HR Q5(b) · Jun 2025 P2H Q2(a)

? What is it?**In plain words**

Draw rays from the **centre** through each corner. The image is **k times** as far along the ray.

$k = \frac{1}{2}$: half as far (smaller). Negative k : the other side of the centre.

Describe: scale factor = $\frac{\text{new length}}{\text{old length}}$;
 centre where the rays meet.

**IT LOOKS LIKE**s.f. $\frac{1}{2}$, centre (1, 2)

describe

How do I do it?**1 Centre**

Mark it.

2 Each corner

Corner – centre (how far across, up).

3 MultiplyBy k ; add back to the centre.**4 Describe**“Enlargement, scale factor k , centre (a, b) ”.**CENTRE + $k \times$ DISTANCE**

The most common variant.

V4 Enlargements

VARIANT 4 · SEE IT DONE, THEN TRY


 Your notes

WHEN YOU SEE → YOU DO

“enlarge ... scale factor $\frac{1}{2}$, centre (1, 2)” → **Half the distance from (1, 2)**

“Describe fully ... A onto B” (bigger or smaller) → **Scale factor and centre**

Negative scale factor → **Through the centre, other side**

WORKED EXAMPLE PAST PAPER

June 2022 · Paper 2HR · Q1(a) · 2 marks

Triangle A: (5, 1), (5, 3), (6, 1). Triangle B: (9, 3), (9, 9), (12, 3). Describe fully the single transformation that maps A onto B.

- | | | |
|---------------|---|------------------------------|
| STEP 1 | height 2 → 6 | ◀ same shape, bigger |
| STEP 2 | $k = 3$ | ◀ scale factor |
| STEP 3 | centre = $(5, 1) - \frac{(4, 2)}{3-1} = (3, 0)$ | ◀ rays back from B through A |
| STEP 4 | Check: $(3, 0) + 3(2, 1) = (9, 3) ✓$ | ◀ check |

ANSWER Enlargement, scale factor 3, centre (3, 0) ✓ two details given ✓

WATCH OUT

- × “enlargement by 3”
- ✓ “scale factor 3, centre (3, 0)”

Without the centre the description scores only 1 mark.

EXAM TIP

Join corresponding corners with long ruled lines — they cross at the centre.

Check yourself

calculator first · answers on page 15

4.1 Jan 2020 P1H

Enlarge (7, 10), (15, 10), (15, 4), (11, 4) by scale factor $\frac{1}{2}$, centre (1, 2).

Answer:

4.2 Nov 2023 P2H

A: (5, 3), (7, 3), (7, 4). B: (1, 5), (7, 5), (7, 8). Describe fully A → B.

Answer:

4.3 Dr Eslam

Enlarge (2, 2), (4, 2), (2, 6) by scale factor $-\frac{1}{2}$, centre (0, 0).

Answer:

V5

VARIANT 5

Combined Transformations

Do them in order on one corner (or the whole shape), then describe the single transformation that does the same.

SEEN IN PAST PAPERS

0 / 18

NOT ASKED YET · BE READY
PRACTICE QUESTIONS: 4

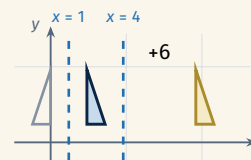

Your notes

? What is it?

In plain words

Do the **first** transformation, then the **second** on the image. Compare the start and the end: what **one** transformation joins them?

Two reflections in parallel lines → a **translation**; in perpendicular lines → a **rotation of 180°** .



IT LOOKS LIKE

reflect in $x = 1$ then $x = 4$

How do I do it?

1 First

Apply it to the corners.

2 Second

Apply it to the new corners.

3 Compare

Start shape and final shape.

4 Describe

The single transformation, fully.

IN ORDER, THEN COMPARE Order matters.

V5 Combined Transformations

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“... then ... Describe the single transformation”

→ **Do both, compare start and end**

“Is Amy correct?” (order swapped)

→ **Do both orders; compare**

Two reflections

→ **Parallel lines: translation; crossing lines: rotation****WORKED EXAMPLE DR ESAM**

Combined · 3 marks

A shape is reflected in the line $x = 1$, then the image is reflected in the line $x = 4$. Describe the single transformation.

STEP 1 $(2, 3) \rightarrow (0, 3)$ ◀ reflect in $x = 1$

STEP 2 $(0, 3) \rightarrow (8, 3)$ ◀ reflect in $x = 4$

STEP 3 $(2, 3) \rightarrow (8, 3) : \begin{pmatrix} 6 \\ 0 \end{pmatrix}$ ◀ compare

STEP 4 Every point moves 6 right (twice the gap between the lines) ◀ all points

ANSWER translation $\begin{pmatrix} 6 \\ 0 \end{pmatrix}$ ✓ same way round, same size ✓

WATCH OUT

× “two reflections”

✓ one single transformation

The question asks for one.

EXAM TIP

Track one corner with coordinates — it is faster than drawing.

Check yourself

calculator first · answers on page 15

5.1 Dr Esam

Rotate 90° clockwise about O , then 180° about O . Single transformation?

Answer:

5.2 Dr Esam

Reflect in the x -axis, then in the y -axis. Single transformation?

Answer:

5.3 Dr Esam

Translate by $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, then by $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$. Single transformation?

Answer:

CHECK YOURSELF

Answers

Every answer below was checked in Python (check_answers.py).

1.1 translation $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$

1.2 (1, 4), (1, 6), (2, 4)

1.3 (-2, 6), (1, 6), (-2, 9)

2.1 (2, 5), (3, 6), (3, 7),
(2, 8)

2.2 (-1, 1)

2.3 (5, -3)

3.1 (3, 6), (1, 6), (0, 5),
(1, 5)

3.2 rotation 180° about
(5, 5)

3.3 (1, -3)

4.1 (4, 6), (8, 6), (8, 3),
(6, 3)

4.2 s.f. 3, centre (7, 2)

4.3 (-1, -1), (-2, -1),
(-1, -3)

5.1 rotation 90°
anticlockwise about O

5.2 rotation 180° about
O

5.3 translation $\begin{pmatrix} 5 \\ -3 \end{pmatrix}$

THE WHOLE TOPIC ON ONE PAGE

Transformations cheat sheet



Your notes
V1 Translation 4/18

$$\begin{pmatrix} 3 \\ -5 \end{pmatrix}$$

Image – start.

V2 Reflection 2/18

$$y = x : (x, y) \rightarrow (y, x)$$

Same distance, other side.

V3 Rotation 4/18

180° about (a, b)

Angle, direction, centre.

V4 Enlargement 8/18centre + $k(\text{corner} - \text{centre})$

Scale factor and centre.

V5 Combined

$$x = 1, x = 4 \rightarrow \begin{pmatrix} 6 \\ 0 \end{pmatrix}$$

In order, then one description.

**TOP 3 LOST MARKS**

1. missing centre or direction × 2. wrong mirror line × 3. giving two transformations ×

BOTTOM LINE Move the corners · describe fully · one transformation only · tracing paper for rotations.